

EXPERIMENT-6

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A) A fruit distribution company records the number of apples and oranges sold each day. For every date, exactly one record exists for apples and one record exists for oranges.

Generate a daily sales comparison report showing the difference between the number of apples sold and the number of oranges sold on the same day. The difference should be calculated as:

Difference = Apples Sold - Oranges Sold

Display the sale date and the calculated difference. The result should be sorted by sale date in ascending order.

```
SELECT
    sale_date,
    SUM(
        CASE
            WHEN fruit = 'apples'
            THEN sold_num
            ELSE -sold_num
        END
    ) AS diff

FROM Sales
GROUP BY sale_date
ORDER BY sale_date;
```

B) A company stores hierarchical employee reporting information in a table. Each record represents a node in a hierarchy where:

- **id** represents the current node.
- **p_id** represents the parent node.

Classify every node into one of the following categories:

- **Root:** A node that does not have any parent.
- **Leaf:** A node that has a parent but does not have any children.
- **Inner:** A node that has both a parent and at least one child.

Display the node ID along with its category. Return the result sorted by node ID.

```
SELECT ID,  
CASE WHEN P_ID IS NULL THEN 'ROOT'  
WHEN ID IN (SELECT P_ID FROM TREE) THEN 'INNER'  
ELSE 'LEAF' END AS TYPE  
FROM TREE;
```